



COMPREHENSIVE DIAGNOSTIC LABORATORY REPORT

Patient Name:	Riya Sharma	Age / Gender:	42 Years / Female
Patient ID:	MC-PT-208451	UHID:	UHID/2026/0071923
Referring Doctor:	Dr. Anjali Kapoor, MD (Internal Medicine)	Lab Accession No:	LAB/2026/091822
Sample Collected:	28-Jul-2026, 07:42 AM	Report Released:	28-Jul-2026, 06:10 PM
Sample Type:	Venous Blood, Random Urine	Billing:	Cashless — Star Health Insurance (Policy No. SH/2024/778102)

1. COMPLETE BLOOD COUNT (CBC) WITH DIFFERENTIAL

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	10.8	g/dL	12.0 - 15.5	L
Total Leukocyte Count (TLC)	11,900	/cumm	4,000 - 11,000	H
Red Blood Cell Count (RBC)	4.1	mill/cumm	3.9 - 5.2	
Haematocrit (PCV)	34.2	%	36.0 - 46.0	L
Mean Corpuscular Volume (MCV)	82.4	fL	80.0 - 100.0	
Mean Corpuscular Hb (MCH)	26.8	pg	27.0 - 32.0	L
MCHC	31.6	g/dL	32.0 - 36.0	L
Red Cell Distribution Width (RDW)	15.1	%	11.5 - 14.5	H
Platelet Count	412,000	/cumm	150,000 - 410,000	H
Neutrophils	68	%	40 - 70	
Lymphocytes	24	%	20 - 40	
Eosinophils	3	%	1 - 6	
Monocytes	4	%	2 - 8	
Basophils	1	%	0 - 1	
ESR (Westergren)	32	mm/hr	0 - 20	H

Interpretation: Mild microcytic-hypochromic anaemia with mildly elevated total leukocyte count and ESR, suggestive of an underlying chronic inflammatory or nutritional deficiency process. Platelet count is marginally elevated (reactive thrombocytosis).



2. KIDNEY FUNCTION TEST (KFT / RENAL PROFILE)

Test	Result	Unit	Reference Range	Flag
Blood Urea	48	mg/dL	15 - 40	H
Serum Creatinine	1.3	mg/dL	0.6 - 1.1	H
Blood Urea Nitrogen (BUN)	22.4	mg/dL	7 - 20	H
Uric Acid	5.6	mg/dL	2.6 - 6.0	
Sodium (Na+)	138	mmol/L	135 - 145	
Potassium (K+)	4.6	mmol/L	3.5 - 5.1	
Chloride (Cl-)	101	mmol/L	98 - 107	
eGFR (CKD-EPI)	61	mL/min/1.73 m ²	> 90	L

Interpretation: Mildly elevated urea and creatinine with reduced eGFR, consistent with Stage 2 chronic kidney disease (mild reduction in renal function). Clinical correlation and repeat testing in 3 months advised.



3. LIVER FUNCTION TEST (LFT)

Test	Result	Unit	Reference Range	Flag
Total Bilirubin	1.4	mg/dL	0.2 - 1.2	H
Direct Bilirubin	0.5	mg/dL	0.0 - 0.3	H
Indirect Bilirubin	0.9	mg/dL	0.2 - 0.9	
SGOT (AST)	58	U/L	8 - 40	H
SGPT (ALT)	64	U/L	7 - 45	H
Alkaline Phosphatase (ALP)	132	U/L	40 - 130	H
Total Protein	7.0	g/dL	6.4 - 8.3	
Albumin	3.9	g/dL	3.5 - 5.0	
Globulin	3.1	g/dL	2.0 - 3.5	
A/G Ratio	1.26	-	1.1 - 2.5	
GGT	51	U/L	9 - 48	H



Interpretation: Mild elevation of transaminases (AST/ALT) and bilirubin suggests mild hepatocellular stress; ultrasound abdomen and correlation with alcohol/medication history recommended.

4. THYROID PROFILE (T3, T4, TSH)

Test	Result	Unit	Reference Range	Flag
Total T3 (Triiodothyronine)	1.1	ng/mL	0.8 - 2.0	
Total T4 (Thyroxine)	5.2	µg/dL	5.1 - 14.1	
TSH (Thyroid Stimulating Hormone)	7.8	µIU/mL	0.4 - 4.5	H
Anti-TPO Antibodies	142	IU/mL	< 34	H

Interpretation: Elevated TSH with positive Anti-TPO antibodies is suggestive of subclinical hypothyroidism, likely of autoimmune (Hashimoto's) origin. Endocrinology consultation recommended.



5. LIPID PROFILE

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	228	mg/dL	< 200	H
Triglycerides	186	mg/dL	< 150	H
HDL Cholesterol	42	mg/dL	> 50	L
LDL Cholesterol	148	mg/dL	< 100	H
VLDL Cholesterol	37.2	mg/dL	5 - 40	
Total Cholesterol / HDL Ratio	5.4	-	< 4.5	H

Interpretation: Dyslipidaemia with elevated LDL and triglycerides and low HDL, indicating increased cardiovascular risk. Lifestyle modification and lipid-lowering therapy to be considered.



6. BLOOD SUGAR PROFILE

Test	Result	Unit	Reference Range	Flag
Fasting Blood Sugar (FBS)	118	mg/dL	70 - 99	H
Post-Prandial Blood Sugar (PPBS)	172	mg/dL	70 - 140	H
HbA1c (Glycated Haemoglobin)	6.4	%	< 5.7	H

Interpretation: FBS, PPBS and HbA1c values are consistent with prediabetes. Dietary counselling, weight management and repeat HbA1c in 3 months advised.

7. URINE ANALYSIS (ROUTINE & MICROSCOPY)

Test	Result	Unit	Reference Range	Flag
Colour	Pale Yellow	-	Pale Yellow	
Appearance	Clear	-	Clear	
pH	6.0	-	4.5 - 8.0	
Specific Gravity	1.018	-	1.005 - 1.030	
Protein	Trace	-	Nil	H
Glucose	Nil	-	Nil	
Ketones	Nil	-	Nil	
Blood	Nil	-	Nil	
Pus Cells (WBC)	4-6	/HPF	0 - 5	
Epithelial Cells	2-3	/HPF	0 - 5	
Red Blood Cells	Nil	/HPF	0 - 2	
Casts	Nil seen	-	Nil	
Crystals	Nil seen	-	Nil	

Interpretation: Trace proteinuria noted, correlating with the mildly reduced renal function seen on the KFT panel; no evidence of active urinary tract infection.



8. CLINICAL NOTES & DOCTOR'S OBSERVATIONS

Patient presented with a 6-week history of generalized fatigue, intermittent joint discomfort, and unintentional weight gain of approximately 2 kg. No history of fever, chest pain, or breathlessness. Patient reports a family history of Type 2 Diabetes Mellitus (mother) and hypothyroidism (maternal aunt). No known drug allergies. Not currently on any regular medication. On examination: BP 132/86 mmHg, Pulse 84/min regular, mild pallor noted, no icterus, no pedal oedema, no palpable lymphadenopathy. Systemic examination was otherwise unremarkable.

Laboratory findings today are notable for mild anaemia, elevated inflammatory markers (TLC, ESR), impaired renal function (Stage 2 CKD range), mild transaminitis, subclinical hypothyroidism with positive autoantibodies, atherogenic dyslipidaemia, and glycaemic parameters in the prediabetic range. These findings together point toward an early metabolic syndrome picture with an autoimmune thyroid component, and warrant a structured follow-up plan rather than any single isolated diagnosis.



9. PROVISIONAL DIAGNOSIS

1. Subclinical Hypothyroidism, likely autoimmune (Hashimoto's Thyroiditis) — Anti-TPO positive.
2. Prediabetes (Impaired Fasting Glucose / Impaired Glucose Tolerance).
3. Mixed Dyslipidaemia with elevated cardiovascular risk.
4. Chronic Kidney Disease, Stage 2 (mildly reduced eGFR) — to be confirmed on repeat testing.
5. Mild Microcytic Anaemia, possibly iron-deficiency related — for further work-up.
6. Mild Hepatocellular Enzyme Elevation — for further evaluation and ultrasound correlation.

10. RECOMMENDATIONS

1. Endocrinology referral for evaluation and management of subclinical hypothyroidism; repeat TSH and free T4 in 6-8 weeks.
2. Nephrology referral advised; repeat Serum Creatinine, eGFR and urine ACR (albumin-creatinine ratio) in 3 months to confirm CKD staging.
3. Initiate lifestyle modification — dietary counselling, structured physical activity (150 min/week), and weight reduction target of 5-7% body weight.
4. Statin therapy to be considered by the treating physician given LDL > 130 mg/dL with additional cardiometabolic risk factors.
5. Iron studies (Serum Iron, TIBC, Ferritin) recommended to evaluate the microcytic anaemia.
6. Ultrasound abdomen (liver, gallbladder, kidneys) recommended to correlate mild transaminitis and renal findings.
7. Repeat HbA1c and fasting lipid profile in 3 months to reassess metabolic status.
8. Follow up with referring physician with all reports for consolidated treatment planning.

Dr. Sameer Nanda
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Consultant Pathologist

Dr. Anjali Kapoor
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Reporting Physician

*** End of Report *** This report has been electronically verified and does not require a physical stamp for validity within the MetroCare Hospital Information System. In case of any discrepancy, please contact the laboratory within 7 days of report generation.